PAGE REPLACEMENT POLICIES
FIFO AND LRU

Subject: Computer Organization and Architecture (COA)

Problem Statement

Consider a system with 3 page frames and the following reference string containing 20 page references. Calculate the total page faults and page hits using FIFO and LRU page replacement policies.

Reference String: 7, 0, 1, 2, 0, 3, 0, 4, 2, 3, 0, 3, 2, 1, 2, 0, 1, 7, 0, 1

Number of page frames: 3

1. FIFO Page Replacement – Solved Table

Replace the page that entered memory first.

Step	Page	Frame 1	Frame 2	Frame 3	Result
1	7	7	–	–	F
2	0	7	–	0	F
3	1	7	0	1	F
4	2	2	0	1	F
5	0	2	0	1	H
6	3	2	3	1	F
7	0	2	3	0	F
8	4	4	3	0	F
9	2	4	2	0	F
10	3	4	2	3	F
11	0	0	2	3	F
12	3	0	2	3	H
13	2	0	2	3	H
14	1	0	1	3	F
15	2	0	1	2	F
16	0	0	1	2	H
17	1	0	1	2	H
18	7	7	1	2	F
19	0	7	0	2	F
20	1	7	0	1	F

Total Page Faults = 15

Total Page Hits = 5

2. LRU Page Replacement – Solved Table

Replace the page that has not been used for the longest time.

Step	Page	Frame 1	Frame 2	Frame 3	Result
1	7	7	–	–	F
2	0	7	–	0	F
3	1	7	0	1	F
4	2	2	0	1	F
5	0	2	0	1	H

6	3	2	0	3	F
7	0	2	0	3	H
8	4	4	0	3	F
9	2	4	0	2	F
10	3	4	3	2	F
11	0	0	3	2	F
12	3	0	3	2	H
13	2	0	3	2	H
14	1	1	3	2	F
15	2	1	3	2	H
16	0	1	0	2	F
17	1	1	0	2	H
18	7	1	0	7	F
19	0	1	0	7	H
20	1	1	0	7	H

Total Page Faults = 12

Total Page Hits = 8

3. Final Comparison

Parameter	FIFO	LRU
Total references	20	20
Number of frames	3	3
Page faults	15	12
Page hits	5	8
Hit ratio	25.00%	40.00%
Fault ratio	75.00%	60.00%

4. Observation

For this reference string, FIFO produces 15 page faults and 5 page hits, whereas LRU produces 12 page faults and 8 page hits. The replacement decision in FIFO is based on arrival order, while LRU is based on recent usage.

Note: The results have been verified using a frame-by-frame simulation of the given reference string.